



Vision Sensor

Sensor head  
IV4-G400CA/IV4-G500CA/IV4-G600CA/  
IV4-G400MA/IV4-G500MA/IV4-G600MA

Sensor Amplifier  
IV4-G120

Instruction Manual

Read this manual before using the product in order to achieve maximum performance. Keep this manual in a safe place after reading it so that it can be used at any time. You can download details of individual functions from the following website:

www.keyence.com/support\_iv4



Symbols

The following symbols alert you to important messages. Be sure to read these messages carefully.

|  |                                                                                                         |
|--|---------------------------------------------------------------------------------------------------------|
|  | Indicates a hazardous situation which, if not avoided, will result in death or serious injury.          |
|  | Indicates a hazardous situation which, if not avoided, could result in death or serious injury.         |
|  | Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.        |
|  | Indicates a situation which, if not avoided, could result in product damage as well as property damage. |
|  | Indicates cautions and limitations that must be followed during operation.                              |
|  | Indicates additional information on proper operation.                                                   |
|  | Indicates tips for better understanding or useful information.                                          |

Safety Information

General Precautions

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"><li>Do not use this product for the purpose to protect a human body or a part of a human body.</li><li>Do not use this product in a hazardous location and/or potentially explosive atmosphere.</li><li>Do not use this product in an application that may cause death, serious injury or serious property damage should a failure with this product occur, such as nuclear power plants, on aircraft, trains, ships, or vehicles, used within medical equipment, playground equipment, roller coasters and other rides, etc.</li></ul>                                                                                                                                                                                                                                                                                      |
|  | <ul style="list-style-type: none"><li>If the product is used in a manner not specified by this manual, the protection provided by the product may be impaired.</li><li>You must perform a sufficient risk assessment for the machine where this product is to be installed prior to installing this product. Provide appropriate protective fail-safe measures on the machine independent from this product in case a failure with this product should occur.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                        |
|  | <p>Before starting or operating the system, check to make sure all system functions are working properly.</p> <ul style="list-style-type: none"><li>KEYENCE never warrants the function or performance of the IV4 Series if it is used in manner that differs from the IV4 Series specifications contained in this instruction manual or if the IV4 Series is modified by yourself.</li><li>When the IV4 Series is used in combination with other instruments, functions and performance may be degraded, depending on operating conditions and the surrounding environment.</li><li>Do not subject the controller or connected devices to a sudden change in temperature. It may cause condensation and may damage instruments or peripherals.</li><li>Remove the power cable from the power supply if you do not use this product for a long time.</li></ul> |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Important Instructions

Observe the following precautions to prevent malfunction and to ensure proper use.

Precautions on use

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>The power of this product and instruments connected to this product must be turned off when the cable is to be installed or removed. Failure to do so may cause an electric shock or product damage.</p> <ul style="list-style-type: none"><li>Use this product with the correct voltage. Failure to do so may cause a malfunction.</li><li>For instructions<ul style="list-style-type: none"><li>Do not turn OFF the power while setting the items or saving the settings. Otherwise, all or part of the setting data may be lost.</li><li>Do not let water, dust or oil stick to the camera/light of the sensor head. Failure to do so may cause a malfunction.</li><li>When this product becomes dirty, do not rub it with a wet cloth, benzene, thinner, or alcohol. Doing so may change the color or shape of the unit.</li><li>If the unit is heavily contaminated, disconnect all the cables including the power supply cable, wipe off the dirt with a cloth soaked with mild detergent, and then wipe with a soft dry cloth.</li></ul></li><li>For external master image registration<p>If the external master image registration is to be performed frequently, set "Write to ROM when using Ext. Master Save?" of the input option to "No" to protect the nonvolatile memory of the internal sensor. When the option is set to "Yes", the nonvolatile memory is guaranteed to write for 100,000 times.</p></li></ul> |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"><li>For automatic focus function<ul style="list-style-type: none"><li>Automatic focus function is used for adjusting the focusing position at the time of installation. Disabled during operation.</li><li>Focusing position can be registered in each program. The program configurations are guaranteed to switch for 100,000 times. If the focusing position does not need to be changed in each program, set [Auto Focus Adjustment Position] to [Common] to extend its duration.</li><li>Do not apply shock or vibration during the focusing position adjustment. Failure to do so may cause a malfunction.</li></ul></li><li>About SD cards<ul style="list-style-type: none"><li>Use a KEYENCE recommended product.</li><li>Close the SD card slot cover while the device is in use.</li><li>When removing the SD card, always perform the [Remove SD Card] procedure before removing the card.</li><li>Do not remove the SD card while it is being accessed. Also do not turn off the system power while the SD card is being accessed. Otherwise, all or part of the setting data may be lost or the memory may be damaged.</li><li>To prevent data loss, periodically backing up to another storage device is recommended.</li></ul></li></ul> |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Measures to be taken when an abnormality occurs

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>In the following cases, turn the power OFF immediately. Using the IV4 Series in an abnormal condition could cause fire, electric shock, or malfunction. Contact your local Keyence representative for repair.</p> <ul style="list-style-type: none"><li>If water or debris enters the IV4 Series.</li><li>If the IV4 Series is dropped or the case is damaged.</li><li>If abnormal smoke or odor emanates from the IV4 Series.</li></ul> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Precautions on installation

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>To use this product correctly and safely, avoid installing it in the following locations. Failure to do so may cause fire, electric shock, or malfunction.</p> <ul style="list-style-type: none"><li>Outdoors</li><li>Altitude above 2500 m</li><li>Locations that are humid, dusty or poorly ventilated</li><li>Locations where the temperature is high such as those exposed to direct sunlight</li><li>Locations where there are flammable or corrosive gases</li><li>Locations where the unit may be directly subjected to vibration or impact</li><li>Locations where water, oil, or chemicals may splash onto the unit</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  | <ul style="list-style-type: none"><li>To improve the anti-noise feature, install the unit following the precautions below. Otherwise, a malfunction may occur.<ul style="list-style-type: none"><li>Ground the FG cable of the sensor amplifier.</li><li>Do not mount the unit in a cabinet where high-voltage equipment is already installed.</li><li>Mount the unit as far from power lines as possible.</li><li>Separate the unit as far as possible from the devices that emit strong electric or magnetic field (such as solenoid or chopper).</li><li>Separate the I/O signal line from the power line or high-voltage line.</li></ul></li><li>For power supply<ul style="list-style-type: none"><li>Noise superimposed on the power supply could cause malfunction. Use a stabilized DC power supply configured with an isolation transformer.</li><li>When using a commercially available switching regulator, be sure to ground the frame ground terminal.</li></ul></li><li>Devices including this unit are precision components. Do not apply shock or vibration.</li><li>When connecting to a network, let engineers who are knowledgeable about networks handle it.</li></ul> |

Precautions on Regulations and Standards

CE and UKCA Markings

KEYENCE Corporation has confirmed that this product complies with the essential requirements of the applicable EU Directive(s) and UK regulations, based on the following specifications. Be sure to consider the following specifications when using this product in the Member States of European Union and in the United Kingdom.

● EMC Directive (CE) and Electromagnetic Compatibility Regulations (UKCA)

- Applicable standard: (BS) EN61326-1, Class A
- Make sure the length of the power cable, I/O cable and Ethernet cable are less than 30 meters long.

Remarks:

These specifications do not give any guarantee that the end-product with this product incorporated complies with the essential requirements of EMC Directive and Electromagnetic Compatibility Regulations. The manufacturer of the end-product is solely responsible for the compliance on the end-product itself according to EMC Directive and Electromagnetic Compatibility Regulations.

CSA Certificate

- This product complies with the following CSA and UL standards and has been certified by CSA.
- Applicable standard: CAN/CSA C22.2 No. 61010-1 UL61010-1

Be sure to consider the following specifications when using this product as a product certified by CSA.

- Use the following power supply.
  - CSA/UL certified power supply that provides Class 2 output as defined in the CEC (Canadian Electrical Code) and NEC (National Electrical Code)
- Connect the sensor amplifier power supply and I/O connector to the same power supply that has Class 2 output.
- Overvoltage category I
- Use this product under pollution degree 3.
- Use this product at the altitude of 2500m or less.
- Indoor use only.

North American Regulations

This product complies with the following North American regulations.

- Applicable regulation : FCC Part15 Subpart B, Class A Digital Device ICES-003, Class A Digital Apparatus
- Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
- FCC CAUTION
  - Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

KC mark (Republic of Korea)

|                                                                          |         |
|--------------------------------------------------------------------------|---------|
| 이 기기는 업무용 환경에서 사용할 목적으로 적합성평가를 받은 기기로서 가정용 환경에서 사용하는 경우 전파간섭의 우려가 있습니다 . | 사용자 안내문 |
|--------------------------------------------------------------------------|---------|

Safety Precautions on LED Product

The degree of risk of this product is shown below.

| Model                                | Light source | Risk Group   |
|--------------------------------------|--------------|--------------|
| IV4-G400CA / IV4-G500CA / IV4-G600CA | White LED    | Risk Group 1 |
| IV4-G400MA / IV4-G500MA / IV4-G600MA | Infrared LED | Exempt Group |

\* LED product is classified as shown below according to IEC 62471.

- Exempt Group
  - Does not pose any photobiological hazard.
- Risk Group 1 (Low-Risk)
  - Does not pose a hazard due to normal behavioral limitations on exposure.
- Risk Group 2 (Moderate-Risk)
  - Does not pose a hazard due to the aversion response to very bright light sources or due to thermal discomfort.
- Risk Group 3 (High-Risk)
  - May pose a hazard even for momentary or brief exposure.

Checking the Package Contents

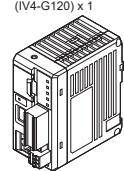
● Sensor head

- Sensor head x 1
  - IV4-G400CA
  - IV4-G400MA
  - IV4-G500CA
  - IV4-G500MA
  - IV4-G600CA
  - IV4-G600MA



● Sensor amplifier

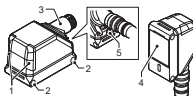
- Sensor amplifier (IV4-G120) x 1



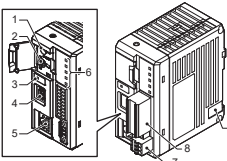
- Instruction Manual (this manual) x 1
- Power terminal block (mounted in the amplifier)
- I/O terminal block (mounted in the amplifier)

Name and Function of Each Part

Name and function of each part of the sensor head

- 
- Status indicators
    - Indicates the operating status of the sensor.
  - Mounting part
    - Used for mounting the sensor head.
    - "Mounting the sensor head" (page 1)
  - Connector for Sensor head/amplifier cable
  - Camera and built-in light
  - Mounting part for the polarizing filter/spread lens attachment

Name and function of each part of the sensor amplifier

- 
- IP reset switch
    - Used for resetting the IP address assigned to this sensor amplifier.
    - For details, refer to "IV4 Series User's Manual."
  - SD card slot
    - Mount the micro SD card.
  - SD card indicator light
    - Indicates the operating status of the micro SD card.
  - Connector for the sensor head/amplifier cable
    - "Cables" (page 2)
  - Ethernet connector
    - Used for connecting to the control panel (IV4-CP70), PC, field network, Ethernet switch or display expansion unit (IV4-DU10). For details of how to connect it, refer to "IV4 Series User's Manual."
  - Indicator light
    - Indicates the operating status of the sensor amplifier.
  - Power terminal block
    - Supplies power to this unit.
  - I/O terminal block
    - Connects to the input and output cables.
  - Connector on the communications unit
    - Used for connecting the communication unit (DL Series). The protection cover is attached during shipment.

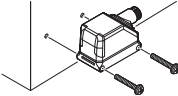
Mounting the sensor head

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Do not place the sensor in an environment that exceeds the limit of the sensor's environmental resistance, or an environment that propagates vibration directly to the sensor. Those may cause a damage or malfunction.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  | <ul style="list-style-type: none"><li>View and optical axis have individual differences. Adjust the sensor position by checking the actual image at the time of installation.</li><li>Place the sensor in a location where ambient light will not affect the image. Ambient light includes solar light, lights of other devices, and photoelectric sensors. Also, be careful when the light intensity of the ambient light changes. Use a shield to protect when the location cannot be changed.</li><li>Place the sensor where no object can block out the internal light or the view of the sensor.</li><li>Detection may become unstable if multiple sensors are placed nearby each other, due to the lighting of each sensor. Take measures as described below.<ul style="list-style-type: none"><li>Delay the timing of external trigger inputs.</li><li>Use a shield to avoid interference.</li></ul></li></ul> |

Mounting the sensor head

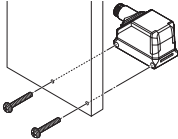
■ When screws are secured from the sensor head

- Screw: M3 x 2
- Tightening torque: 0.3 to 0.6 N·m



■ When screws are secured from the wall

- Screw: M4 x 2
- Tightening torque: 0.7 to 1.5 N·m
- Screw hole on the sensor head: M4 (screw depth : 3.5 mm)

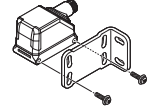


Attaching the optional mounting bracket

■ Vertical mounting bracket (OP-87908)

- Attach the bracket using the screws attached to OP-87908.
- Tightening torque: 0.7 to 1.5 N·m

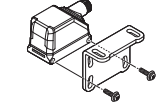
Mounting examples



■ Transverse mounting bracket (OP-87909)

- Attach the bracket using the screws included with the adjustable bracket (OP-87909).
- Tightening torque: 0.7 to 1.5 N·m

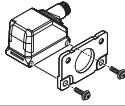
Mounting examples



■ Adjustable bracket (OP-87910)

1 Attach the bracket using the screws included with the adjustable bracket (OP-87910).

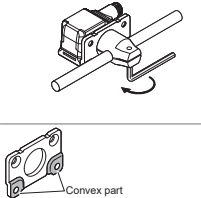
- Tightening torque: 0.7 to 1.5 N·m
- Mounting examples



- Connect the convex parts to the sensor head.

2 Attach and fix the bracket on the brace side.

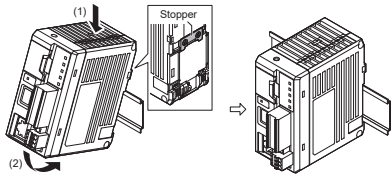
- Tightening torque : 5 N·m
- Mounting examples



Mounting the Sensor Amplifier

|  |                                                                                                                                                                                                                                                                                                               |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Mount the sensor amplifier as follows for heat release.</p> <ul style="list-style-type: none"><li>For the mounting direction, place the indicator light toward the front.</li><li>Provide the sensor with space of 30mm or more on all sides.</li><li>Set it on DIN rail mounted on sheet metal.</li></ul> |
|  | <p>A: 30 mm or more</p>                                                                                                                                                                                                                                                                                       |

1 Fit the stopper on the back of the sensor amplifier (center side) to the DIN rail (1), and then push the sensor amplifier in on the arrow mark direction (2) until you hear the clicking sound.

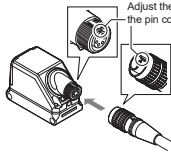


When detaching the sensor amplifier, push the upper part down to the front while pushing the sensor amplifier in the reverse direction of the (1) arrow mark (top).

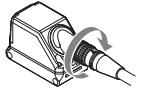
## Cables

### Connecting the sensor head/amplifier cable and sensor head

1 Connect the cable to the sensor head. 2 Tighten the connector by turning the screw on the connector in the clockwise direction.



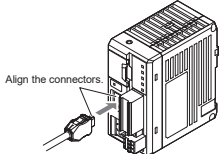
Tightening torque: 0.6 to 0.8 N·m



**NOTICE** To tighten the screw by hand, turn it until the connector of the cable touches the sensor head firmly. Then turn it more about by 5° using a tool such as pliers.

### Connecting the sensor head/amplifier cable and sensor amplifier

Align the connectors and insert the cable until a clicking sound is made.

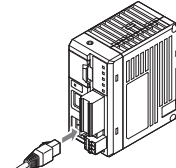


To disconnect the cable, push in the tabs on both sides of the connector and pull out the connector.

### Connecting the cable to the sensor amplifier

#### ■ Connecting the Ethernet cable or LAN cable

Connect the cable to the sensor amplifier as shown below.

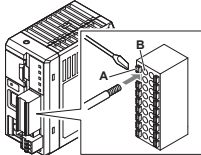


To disconnect the cable, pull the cable while pushing in the tab on the Ethernet cable.

#### ■ Wiring the I/O terminal block or LAN cable

Insert the wire to B by pushing A in with a screwdriver.

Pull the wire out slightly to confirm that it is secured.

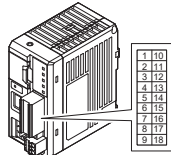


**CAUTION** Use a wire with 60°C or higher temperature rating as the electrical wire.

**NOTICE** Follow the instructions shown below to avoid damage caused by bad connections.

- The nominal cross-sectional area of the wire for connecting the I/O terminal block should be 0.2 mm<sup>2</sup> to 1.5 mm<sup>2</sup> (AWG16 to 26).
- The stripped part of the wire should not be soldered (pre-soldered).
- The length of the insulation removed from the wire should be around 10mm.
- Insert the wires all the way into the power terminal block.
- Wire the power cables directly or using bar terminals into the terminal block.

#### ● Connector pin layout



| Terminal No. | Name    | Assigning default value | Description                                                                                                                                                                                |
|--------------|---------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | IN1     | Ext. Trigger ↑          | Set external trigger.<br>Rising timing (↑) or falling timing (↓) can be set.                                                                                                               |
| 2            | IN2     | OFF                     | Input assignable function<br>• Program bit0 to bit6<br>• Clear Error<br>• Ext. Master Save<br>• SD Card Save Stop<br>• OCR Sequential Number Count-up<br>• Count Reset<br>• OFF (not used) |
| 3            | IN3     | OFF                     |                                                                                                                                                                                            |
| 4            | IN4     | OFF                     |                                                                                                                                                                                            |
| 5            | IN5     | OFF                     |                                                                                                                                                                                            |
| 6            | IN6     | OFF                     |                                                                                                                                                                                            |
| 7            | IN7     | OFF                     |                                                                                                                                                                                            |
| 8            | IN8     | OFF                     |                                                                                                                                                                                            |
| 9            | Unused  | Unused                  | Unused                                                                                                                                                                                     |
| 10           | OUT1    | Total Status OK (N.O.)  | Output assignable function<br>• Total Status OK/NG                                                                                                                                         |
| 11           | OUT2    | BUSY (N.O.)             | • RUN                                                                                                                                                                                      |
| 12           | OUT3    | Error (N.C.)            | • BUSY                                                                                                                                                                                     |
| 13           | OUT4    | OFF                     | • Error                                                                                                                                                                                    |
| 14           | OUT5    | OFF                     | • SD card error                                                                                                                                                                            |
| 15           | OUT6    | OFF                     | • Status result of each tool                                                                                                                                                               |
| 16           | OUT7    | OFF                     | • Logical operation result of each tool                                                                                                                                                    |
| 17           | OUT8    | OFF                     | • Type discrimination result                                                                                                                                                               |
| 18           | OUT COM | OUT COM                 | • Master judgment result<br>• OFF (not used)                                                                                                                                               |

Compatible cable specification: AWG16 to 26

**Reference** The output assignment, N.O./N.C., and input line assignment can be changed. Details of the functions that can be assigned other than the above, refer to "IV4 Series User's Manual."

**NOTICE** For input cables of this sensor, connect with non-contact output (transistor output/SSR output). For contact output (relay output), incorrect input may occur due to contact bouncing.

#### ■ Supplying power to the sensor amplifier

The power terminal block is mounted on the front of the sensor amplifier.

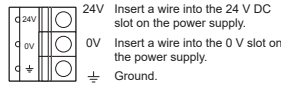
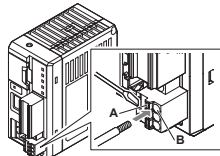
**CAUTION** Use a wire with 60°C or higher temperature rating for the power source.

**NOTICE** Follow the instructions shown below to avoid damage caused by bad connections.

- The nominal cross-sectional area of the wire for connecting the power should be 0.8 mm<sup>2</sup> to 1.3 mm<sup>2</sup> (AWG16 to 18).
- The length of the insulation removed from the wire should be around 9 mm.
- The stripped part of the wire should not be soldered (pre-soldered).
- Insert the wires all the way into the power terminal block.
- Wire the power cables directly or using bar terminals into the terminal block.

#### 1 Insert the wire to B by pushing A in with a screwdriver.

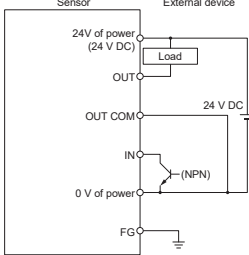
Pull the wire slightly to confirm that it is secured.



#### ■ Connection diagram

##### ● Selecting NPN output

When NPN is selected as the Polarity



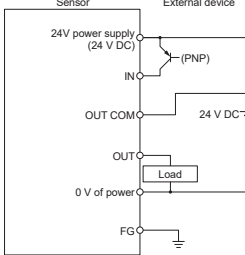
**Point** OUT COM is a common terminal for OUT. Be sure to connect it to 0V of the power supply.

**Reference** For the electrical specifications of the input/output circuit, refer to "Sensor amplifier" (page 2).

- 0 V (power supply) and FG are insulated.
- Use OUTCOM and 0 V of power with a potential difference of 26.4V or low.

##### ● Selecting PNP output

When PNP is selected as the Polarity



**Point** OUT COM is a common terminal for OUT. Be sure to connect it to 24V of the power supply.

## Specifications

### Sensor head

| Model                    |                               | IV4-G400CA/IV4-G400MA/IV4-G500CA/IV4-G500MA/IV4-G600CA/IV4-G600MA |
|--------------------------|-------------------------------|-------------------------------------------------------------------|
| Power supply             | Average power consumption     | 2.5 W (without AI lighting unit)<br>6.6 W (with AI lighting unit) |
|                          | Operating ambient temperature | 0 to +50°C (No freezing) <sup>1</sup>                             |
| Environmental resistance | Operating ambient humidity    | 85% RH or less (No condensation)                                  |
|                          | Enclosure rating <sup>2</sup> | IP67                                                              |

<sup>1</sup> If the operating ambient temperature exceeds 40°C, confirm that the case temperature does not exceed the rated 65°C by appropriate heat dissipation measures after installation as specified by KEYENCE.

Measure the case temperature at the side marked "IV4 Series".

<sup>2</sup> Except when the polarizing filter (OP-88642/OP-88643/OP-88646/OP-88647) or the spread lens attachment (OP-88909) is mounted.

### Sensor amplifier

| Model                    |                                                                                                                                                                                                                                                                                                               | IV4-G120                                                                                                                                                                                                                                |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input                    | Switchable between non-voltage input and voltage input<br>For non-voltage input: ON voltage 2 V or lower, OFF current 0.1 mA or lower, ON current 2 mA (short circuit)<br>For voltage input: Maximum input rating 26.4 V, ON voltage 18 V or higher, OFF current 0.15 mA or lower, ON current 2 mA (for 24 V) |                                                                                                                                                                                                                                         |
|                          | Number of inputs                                                                                                                                                                                                                                                                                              | 8 (IN1 to IN8)                                                                                                                                                                                                                          |
| Output                   | Open collector output NPN/PNP is switchable, N.O./N.C. Is switchable.<br>Maximum rating 26.4 V 50mA, residual voltage 1.5V or less <sup>1</sup>                                                                                                                                                               |                                                                                                                                                                                                                                         |
|                          | Number of outputs                                                                                                                                                                                                                                                                                             | 8 (OUT1 to OUT8)                                                                                                                                                                                                                        |
| Power supply             | Power voltage                                                                                                                                                                                                                                                                                                 | 24 V +25%,-20% (including ripple)                                                                                                                                                                                                       |
|                          | Current consumption                                                                                                                                                                                                                                                                                           | 2.2A or less (at 19.2V), 1.7A or less (at 24.0V) (without AI lighting unit, and including an output load of 120 mA)<br>4.0A or less (at 19.2V), 3.2A or less (at 24.0V) (with AI lighting unit, and including an output load of 120 mA) |
|                          | Average power consumption                                                                                                                                                                                                                                                                                     | 9.5 W (sensor amplifier only, no output load)                                                                                                                                                                                           |
| Environmental resistance | Operating ambient temperature                                                                                                                                                                                                                                                                                 | 0 to +50°C (No freezing) <sup>2</sup>                                                                                                                                                                                                   |
|                          | Operating ambient humidity                                                                                                                                                                                                                                                                                    | 85% RH or less (No condensation)                                                                                                                                                                                                        |

<sup>1</sup> Make sure to keep a total of each output with 160mA or less. Residual voltage is specified at an output current of 20 mA or less. The output current is 2.0 V at 50 mA.

<sup>2</sup> If the operating ambient temperature exceeds 40°C, confirm that the case temperature does not exceed the rated 65°C by appropriate heat dissipation measures after installation as specified by KEYENCE.

Measure the case temperature at the metal on the back of the sensor amplifier.

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A4SWM1-MAN-2033

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17958GB 2034-1 96M17958 Printed in Japan